EE 2300 DESIGN: FREQUENCY RATE PULSER

Build a circuit that takes a single audio frequency input and flashes three LEDs of different colors corresponding to the sound level at three different frequencies. For example, you might choose a blue LED that corresponds to low frequencies, a red LED for medium frequencies, and a yellow LED for high frequencies. The LEDs will flash with the beat of the signal, meaning that when the sound at a specific frequency is loud enough, the LED will flash and the flash should be bright enough and last long enough to be clearly noticeable.

Design Requirements

- You may choose whatever frequencies you like for the three LEDs, but a good starting set might be 100 Hz for the low frequency, 1,000 Hz for the medium frequency, and 3,000 to 5,000 Hz for the high frequency. It is not recommended to go much higher than 5,000 Hz.
- The voltage threshold level for when the LED turns on is up to you, however, it should be set so that the LEDs will flash sometimes, but will not be on all of the time.
- You can use up to two DC power supplies to power your circuit.

Comments

- Clearly, you will need three filter circuits to sort out the different frequencies. You do not need to look at all frequencies, just a set that is representatively of low, middle, and (sort of) high frequencies. You can design the filters however you want, but bandpass filters with a reasonably high Q might be a good starting point.
- Presumably, the source will be from a phone or computer or MP3 player. You will almost certainly have to amplify the signal before passing it to the filters. (Or as part of the filters, if using active filters.)
- To make the LED flash more sustained and more noticeable, you may want to use a peak detector together with a comparator to sustain the length of the flash.
- You may want to make some aspects of the circuit adjustable using potentiometers. For instance, making the switching threshold for any comparators or the gain for any amplifiers adjustable will help in tweaking the operation of the circuit.
- NOW HEAR THIS! If you connect your phone or other expensive gadget to an active circuit, you must double- and triple-check connections before hooking your source in. Make sure that there are no DC power supply voltages accidentally attached at the input. A 10-V supply attached to the audio jack of your player will probably cause permanent damage.
- As always, take a “divide and conquer” approach. Start with one filter at one frequency. Build it and measure its frequency response. Then add whatever amplification may be needed. Finally, build and test a simple sample-and-hold comparator (peak detector) that will drive an LED. Once the pieces are working, hook it all together to make one complete “channel”. When you have one channel working to your satisfaction, repeat for the other two channels.

Testing / Reporting

- Your lab supervisor will test your circuit. Be prepared to show the operation of your circuit with a sound source of your choosing.
- You should measure the frequency response of each of the three filters that you used in your design. These plots will be included in your lab report.
- Write a short report that includes:
    - Complete circuit diagram
    - Photograph of your circuit
    - Written description of the design of the circuit.
    - Any additional discussion about the performance of your circuit, or challenges encountered during the design.